



MATHEMATICS

0580/22

Paper 2 (Extended)

May/June 2018

MARK SCHEME

Maximum Mark: 70

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfw	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1	2 [h] 55 [min]	1	
2	$7x - 56$ final answer	1	
3	$[a =] 15$ $[b =] -27$	2	B1 for each or SC1 for reversed answers
4(a)	$[w =] 7$	1	
4(b)	$[12x =] 36$	1	
5	24	2	B1 for 17 or 41 identified
6	$\frac{8}{12}$ and $\frac{1}{12}$ oe	M1	For correct fractions with a common denominator $12k$
	$\frac{7}{12}$ cao	A1	
7	320	2	M1 for $180 + 140$ oe
8(a)	1.36×10^6 oe	1	
8(b)	5.21×10^{-3} oe	1	
9	Correct perpendicular bisector of AB with 2 pairs of correct arcs	2	B1 for correct perpendicular bisector of AB with no or wrong arcs or for 2 pairs of correct arcs
10	$(x + 2)(y + 3)$ final answer	2	B1 for $y(x + 2) + 3(x + 2)$ or $x(y + 3) + 2(y + 3)$
11	80	2	M1 for $\left(\frac{12}{3}\right)^2$ or $\left(\frac{3}{12}\right)^2$ oe or $\frac{3^2}{5} = \frac{12^2}{A}$ oe
12	7 cao nfw	2	B1 for $31 + 0.5$ or $5 - 0.5$ or 31.5 or 4.5 seen
13	15 and 22	2	M1 for 1.5×10 or 1.1×20
14	62	3	M1 for [height =] $21 \div 7$ M1 for $2(1 \times their3 + their3 \times 7 + 1 \times 7)$ oe

Question	Answer	Marks	Partial Marks
15	628 or 628.3 to 628.4 cm^3	3	B2 for 628 or 628.3 to 628.4 or M1 for $5^2 \times 8 \times \pi$ B1 for cm^3
16	7.5 nfw	3	M2 for $[OB^2 =] \left(\frac{12}{2}\right)^2 + 4.5^2$ oe or B1 for recognition of right angle
17	30	3	M2 for $\frac{1}{2}(8+2) \times v [= 150]$ oe or M1 for $\frac{1}{2} \times 6 \times v$ or $2 \times v$ oe
18(a)	$d = 4.9t^2$	2	M1 for $d = kt^2$
18(b)	19.6	1	FT their 4.9×4
19	$y > 2$ oe final answer $y \geq 3 - x$ oe final answer	3	B1 for $y > 2$ oe final answer B2 for $y \geq 3 - x$ oe final answer or B1 for $y = 3 - x$ oe soi or SC2 for $y \geq 2$ oe and $y > 3 - x$ oe final answer
20(a)	C^2	2	B1 for any correct matrix calculation evaluated
20(b)	-9	1	
20(c)	The determinant is 0 oe	1	e.g. it is singular.
21(a)	140 000	1	
21(b)	Points correctly plotted at (40, 80) and (80, 150)	1	
21(c)	Correct ruled line of best fit	1	
21(d)	80 000 to 110 000	1	FT their straight line provided it has positive gradient
22(a)	$6a - 2b$ or $2(3a - b)$	2	M1 for $4a + b - (-2a + 3b)$ or better
22(b)	$5a - b$	2	M1 for a correct route e.g. $\overrightarrow{OD} + \overrightarrow{DE}$, $4a + b + a - 2b$, $\overrightarrow{OE}$
23(a)	5	3	M2 for $20 - x + x + 8 - x = 23$ or better or B1 for identifying the correct region $A \cup B$
23(b)	$\frac{7}{30}$ oe	2	B1 for $\frac{7}{c}$ or $\frac{k}{30}$

Question	Answer	Marks	Partial Marks
24(a)	$\frac{4}{5}$ oe	2	M1 for $\frac{2}{3} \times p = \frac{8}{15}$ or better
24(b)	$\frac{1}{15}$ oe	3	3FT $(1 - \text{their } \frac{4}{5}) \times \frac{1}{3}$ correctly evaluated M2 for $(1 - \text{their } \frac{4}{5}) \times (1 - \frac{2}{3})$ oe or M1 for $1 - \text{their } \frac{4}{5}$ or $1 - \frac{2}{3}$
25(a)	$[y =] -\frac{2}{5}x + 3$ or $[y =] -0.4x + 3$ final answer	4	B2 for [gradient of perpendicular =] $-\frac{2}{5}$ oe or M1 for [gradient =] $\frac{24-9}{22-16}$ or $-\frac{22-16}{24-9}$ M1 for substituting (5, 1) into $y = \text{their } mx + c$
25(b)	(20, 19)	2	M1 for $\frac{2}{3}(22-16)+16$ or $\frac{2}{3}(24-9)+9$ oe or SC1 for answer (18, 14)